PREDICTING LINEAR RELATIONAL DECODABILITY WITHOUT THE RELATION

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ABSTRACT

Linear relational embeddings (LREs) approximate a language model's subject-to-object mapping with an affine function. Their faithfulness varies enormously across relations – near zero for PERSON_FATHER, at ceiling for COUNTRY_CAPITAL_CITY. We investigate the factors that contribute to this variation and demonstrate that it can be significantly predicted from a readout that is not aware of the underlying relationship. To do so, we propose a novel form of the Jacobian lens, the *contrastive Jacobian lens*: instead of model outputs for "France", we instead contrastively predict outputs for "France" that are not also outputs for "Spain". After estimating this contrastive Jacobian lens, we measure how highly it ranks the model's answer for a relation – without using any relation-specific information in estimation and prediction. Across 28 relations of the LRE dataset on Olmo-3-7B this relation-agnostic score correlates with LRE faithfulness at $\rho = 0.77$ ($p < 10^{-5}$). This suggests that linear decodability is largely a property of what a subject's representation already carries, rather than of the relation being decoded. After attempting to move the unit of analysis from the relation to the individual fact, we conclude that new relation datasets and analysis methods are needed to address several problems in existing datasets, most prominently semantically weak first tokens (e.g., most predicted universities' first token is "University") and subject ambiguity (e.g., the LLM associates obscure actor names with non-actor entities).

1 INTRODUCTION

Hernandez et al. (2024) showed that a language model's mapping from a subject to the object of a relation is often well approximated by an affine function of the subject's hidden state – a *linear relational embedding* (LRE). They also showed that this works far better for some relations than others. On Olmo-3-7B, we reproduce the same spread: an LRE fitted on COUNTRY_CAPITAL_CITY agrees with the model's own prediction 75% of the time while one fitted on PERSON_FATHER agrees 9% of the time.

Why? The natural reading is that some relations are "more linear" than others – that linearity is a property of the relation. We report evidence that it is better understood as a property of the *subject's representation*. Concretely, we show that one can predict how faithful an LRE will be *before fitting it*, and without using the relation at all.

Building on recent work by Gurnee et al. (2026), we introduce the *contrastive Jacobian lens*, which evaluates the Jacobian at individual reference entities and applies it to *differences* between entity representations: *France* minus *Spain* rather than *France* alone. What the subtraction leaves is what is specific to this entity among others of its kind, and the Jacobian carries that residual forward to the vocabulary. The construction changes what the readout is about: from a category to an individual. The question it answers – *what is this model ready to say about this entity, as against others of its kind?* – is not specific to relations.

We estimate the contrastive Jacobian lens under a single neutral prompt shared by every relation, at one layer fixed in advance, and ask how highly it ranks the model's own answer. The lens sees no relation template, no objects and no (subject, object) pairs, yet its score correlates with LRE faithfulness across 28 relations at $\rho = 0.77$.

We make four contributions. (i) We introduce the contrastive Jacobian lens, a supervision-free readout of what a model is disposed to say about an individual entity. (ii) We turn it into a relation-agnostic diagnostic of linear decodability and show that it predicts LRE faithfulness. (iii) We show the prediction holds up under leave-one-domain-out and under controls for object cardinality and answer-token frequency. (iv) After attempting to move the unit of analysis from the relation to the individual fact, we provide an analysis of several problems in existing datasets – most prominently semantically weak first tokens and subject ambiguity – that hinder further progress on understanding LREs in language models.

2 Related work

Linear relational embeddings. Hernandez et al. (2024) brought the idea of modeling relations as linear maps between concept representations (Paccanaro & Hinton, 2000) to transformer language models (Vaswani et al., 2017). Rather than training such maps, they estimate them from a trained model by approximating the mapping from a subject's hidden state to the model's output with an affine function derived from the model's Jacobian. The approximation is faithful for some relations and poor for others. Merullo et al. (2025) trace part of this variation to pretraining data: LREs are more faithful when subjects and objects frequently co-occur in the corpus (we compare against such frequency measures in Appendix A.1). A more fundamental question is whether linear structure reflects what a model has learned at all, since many parametrizations induce the same next-token distribution. Marconato et al. (2024) refine the notion of relational linearity and show that, under suitable conditions, it holds either in all models that induce the same next-token distribution or in none. Valer et al. (2026) build on this formulation with an empirical probe based on the Kullback–Leibler divergence and find that relational linearity varies across models and layers and with how the relation is phrased. These works evaluate and explain linearity at the level of the relation. We ask instead how much of it can be predicted from how the model represents the subject, using a readout estimated without the relation or any of its objects. We also identify object cardinality as a second factor that shapes LRE faithfulness (§8).

Subject representations in factual recall. Mechanistic studies converge on the view that much factual knowledge is attached to the subject before the relation is processed. Feed-forward layers operate as key–value memories whose keys respond to patterns in the input and whose values induce distributions over the output vocabulary (Geva et al., 2022), and causal tracing localizes factual associations in middle-layer MLPs at the last subject token, where targeted edits can change individual facts (Meng et al., 2022). Building on these findings, factual recall has been decomposed into stages: MLP layers enrich the subject representation with many of its attributes, the relation is propagated to the final position, and attention heads extract the requested attribute (Geva et al., 2023; Nanda et al., 2024). Nanda et al. (2024) further report that the lookup performed by early MLPs depends largely on the tokens of the subject's name rather than on the preceding context. Gottesman et al. (2026) show that knowledge acquisition tracks how often subject and answer co-occur, though frequency alone does not fully explain learning.

How facts are stored. Theoretical work offers two accounts of how transformers store facts. Nichani et al. (2025) model attention and MLP weights as associative memories and prove that a one-layer transformer can store a number of facts that scales linearly with its parameter count, with storage traded off between self-attention and MLP parameters. Ravfogel et al. (2026) propose a geometric alternative: in a controlled one-layer setting, subject embeddings encode linear superpositions of their associated attributes, and a small MLP acts as a relation-conditioned selector through ReLU gating rather than as a key–value lookup. The geometric account is close in spirit to our hypothesis. If an entity's attributes are already superposed in its representation, a relation only needs to select among them, and selection should succeed to the extent that the attribute is prominently encoded.

Lenses. Lenses read out intermediate hidden states in terms of the output vocabulary and differ mainly in the map they apply: the model's own unembedding (nostalgebraist, 2020), an affine translator learned per layer (Belrose et al., 2023), or the model's input–output Jacobian averaged over a large corpus and applied to raw activations, with the aim of identifying representations the model is poised to verbalize (Gurnee et al., 2026). Our contrastive variant evaluates the Jacobian at reference entities of the same type and applies it to differences between entity representations. This subtracts

what the reference entities share, so that the top-ranked tokens describe an individual entity rather than its category. As a predictor of LRE faithfulness, the non-contrastive Jacobian lens performs comparably (Appendix A).

3 BACKGROUND

An LRE for relation r approximates the model's computation as

$$o \approx \beta W s + b_\beta, \tag{1}$$

where s is the hidden state at the subject's final token and o the output state. Write f for the model's own map between the two: f sends the subject's hidden state at an intermediate layer ℓ to the final-layer state at the last position. We average local first-order approximations of f over a small set $\{s_i\}$ of reference subjects and scale their subject-dependent terms by β:

$$W = \mathrm{mean}_i\, J_i, \qquad b_\beta = \mathrm{mean}_i[\, f(s_i) - \beta J_i s_i \,], \tag{2}$$

where J_i is the Jacobian of f at s_i. At $\beta = 1$, this is the averaged first-order approximation (Hernandez et al., 2024).

The scalar β corrects a systematic weakness of the first-order estimate. A Jacobian taken at a reference subject describes the model's map only in a neighborhood of that subject; extrapolated to a different subject it points in roughly the right direction but is too short. This matters because the prediction decomposes into a subject-dependent term and a constant: expanding b_β gives

$$\beta W s + b_\beta \;=\; \mathrm{mean}_i[\, f(s_i) + \beta J_i (s - s_i) \,], \tag{3}$$

in which only the second term depends on which subject is being decoded. Left unscaled, the shared anchor $\mathrm{mean}_i\, f(s_i)$ dominates and the operator returns much the same object for every subject. Setting $\beta > 1$ amplifies the subject-specific contribution relative to that anchor, and so does change which object is ranked first. Hernandez et al. (2024) fit β per relation alongside the layer; we follow their protocol and select it from $\{0.5, 1, 1.5, 2, 3, 5\}$.

Faithfulness is the rate at which decoding $\beta W s + b_\beta$ over the full vocabulary reproduces the *model's own* top-1 prediction (not the gold object). It is therefore a measure of how well an affine map captures what the model does. Faithfulness defined this way is the quantity we predict.

4 THE CONTRASTIVE JACOBIAN LENS

Gurnee et al. (2026)'s Jacobian lens is broadly similar to LRE. But instead of a relation-specific prompt and a focused target position (the object), it computes the average $W = \mathrm{mean}_i\, J_i$ over many diverse prompts and many output positions. The Jacobian lens also does not use the "anchor", the zero-order term of the Taylor expansion; so it is simply $W = \mathrm{mean}_i\, J_i$.

We derive our *contrastive Jacobian lens* from Eq. 3, restated as:

$$f(s) \approx \beta W s + b_\beta \;=\; \mathrm{mean}_i[\, f(s_i)] + \beta\, \mathrm{mean}_i[\, J_i (s - s_i) \,] \tag{4}$$

Following Gurnee et al. (2026), we also drop the anchor term $\mathrm{mean}_i[\, f(s_i)]$; see discussion in §10. This also means that we do not need β since the scaling does not change the output distribution over tokens. The final form of the contrastive Jacobian lens is then:

$$\mathrm{CJL}(s) = \mathrm{mean}_i[\, J_i (s - s_i) \,] \tag{5}$$

Note that CJL takes into account the difference between s and the reference point s_i – as it should if we want to approximate $f(s)$ using the first derivative. This difference formalization is also the main motivation for introducing our alternative lens. We are not interested in what the model associates with an entity like "Russia" in general (e.g., "Europe"). Instead we are interested in what the model associates with "Russia", but not with other reference entities like "Spain" and "Italy". Examples of such contrastive associations or "Russia" are "vodka" and "Kremlin". The basic idea we will investigate is that such top-ranked contrastive associations give rise to linearity.

To instantiate the contrastive Jacobian lens for factual relations, we need (i) a prompt that evokes factual associations and (ii) a reference set that contrasts an entity with entities of the same type.

As the prompt we adopt `"{} is known for"`. This a neutral prompt, i.e., it is not customized for any specific factual relations. We use it for all our experiments unless otherwise stated.

As the reference set, we use entities of the same type as the subject, e.g., other countries if the subject is a country, other landmarks if the subject is a landmark etc. The non-contrastive standard Jacobian lens applied to an input s outputs anything the model is likely to say in a context in which s occurs, including features it shares with other same-type entities. The contrastive Jacobian lens is trained to output only what differentiates s from other same-type type.

Computation. We do not compute J_i explicitly, but obtain each Jacobian–vector product $J_i u$ by the double-backward construction (Pearlmutter, 1994), at the cost of a constant number of forward and backward passes; this is what makes Eq. 7 affordable at $d_{\text{model}} = 4096$ (Appendix B).

The relation score. We decode CJP(s) over the full vocabulary and record the rank of m_e, the model's own top-1 answer for relation r, obtained under the relation's in-context prompt (§5). Writing $\mathrm{rk}_{\mathrm{CJL}(e)}(m_e)$ for that rank, the diagnostic for a relation is

$$A_r \;=\; -\operatorname{median}_e \log_{10} \mathrm{rk}_{\mathrm{CJL}(e)}(m_e), \tag{6}$$

so higher A_r means the lens agrees with the model more.

What makes A_r relation-agnostic. Nothing on the lens side of Eq. 7 is specific to r: the prompt is neutral and shared, the layer is fixed at 17 for all relations (Appendix C), and the differences are subject-minus-subject – *France* minus *Spain*, never *France* minus *Paris* – so no (subject, object) pair of the relation is used in estimation. Relation-specificity enters in exactly one place, the *target* m_e, the quantity faithfulness is defined against.

5 Experimental setup

Model. Olmo-3-7B, in `float32` with eager attention, as the Jacobian–vector products require both (Appendix B).

Relations. The LRE dataset (Hernandez et al., 2024) contains 47 relations. Our prompt presupposes an entity subject, so we fixed a subject-type filter in advance using the dataset's declared `domain_name` field: the 28 relations with named-entity subjects (*person*, *country*, *name*, *company*, *landmark*, *superhero*, *city*, *pokemon*, *product*, *star*). Excluded are relations over words as words (*verb*, *adjective*, *word*) and over abstracts (*occupation*, *task*, *characteristic*, *degree*), for which `"{} is known for"` is a category error. No other relation was excluded.

Measurement. Both quantities are read off the same prompt, so that they are comparable. Following Hernandez et al. (2024), a relation is presented to the model *in context*: eight (subject, object) examples of that relation are concatenated, one per line, and the query subject is appended without its object, so that the model's next token is its answer. This is what defines "the model's own prediction" for both measures. The subjects of a relation are shuffled once and partitioned into three disjoint sets: eight in-context examples, eight reference subjects from which W and b_β are estimated, and the remainder as scored test subjects.

Faithfulness is then recomputed for all 28 relations under this protocol. Layer and β are chosen on a 6×6 grid – six candidate layers for the subject state $(5, 11, 14, 17, 20, 23)$ crossed with six values of β $(0.5, 1, 1.5, 2, 3, 5)$ – by the mean over the test subjects. Hernandez et al. (2024) likewise fit both per relation; the six layers span the network at roughly even spacing and the β range brackets the values they report.

The lens scores 11–20 subjects per relation, depending on how many remain once the in-context and reference sets are removed. For the results in §6, the lens and the LRE sample their splits independently, so the particular subjects entering each set differ between the two measurements. Independent splits introduce sampling variation into this comparison. Because the LRE hyperparameters are selected on the scored subjects, its reported faithfulness is an in-sample estimate; these results do not establish performance on a separate held-out evaluation set.

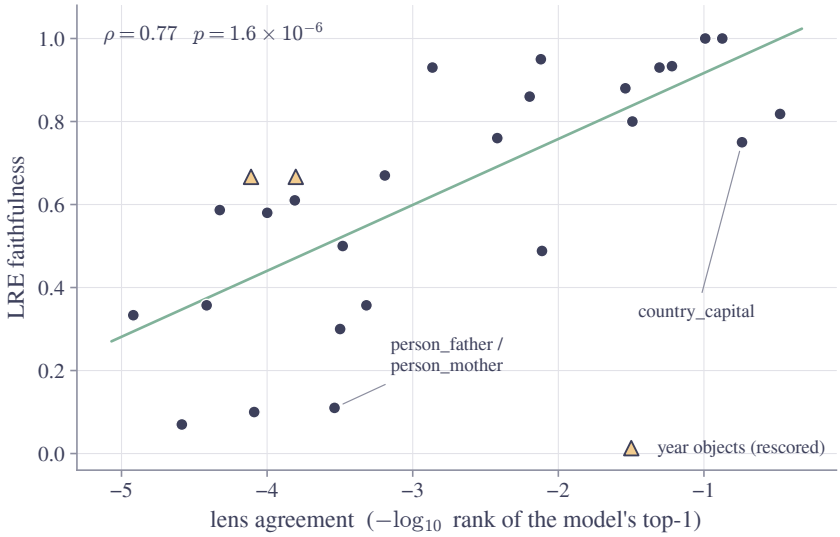


Figure 1: LRE faithfulness against the relation-agnostic lens score, 28 relations. The lens sees no relation template, no objects, and no (subject, object) pairs, and uses one fixed layer throughout. Triangles mark the two relations with numeric objects, rescored as described in Appendix D.

Crucially, the in-context prompt is used only to obtain the *target*; the lens itself is always computed under the neutral prompt of §4.

6 RESULTS

1 <<< hs: fig:main: ideally, all relation points that are part of the discussion (e.g., second half of section6) should be labeled, the same way that country capital and personmother/father are >>>

2 <<< hs: is the labeling of person mother/father correct? i think not: we need two labels pointing to two different points >>>

Figure 1 shows the result. The relation-agnostic score predicts LRE faithfulness at Spearman $\rho = 0.771$ ($p = 1.6 \times 10^{-6}$).

The correlation is not carried by cardinality. Relations differ greatly in how many distinct objects they have; k, the mean number of subjects per object, ranges from 1.0 to 189. High k should make faithfulness easier, and k is indeed associated with faithfulness on its own, though only weakly ($\rho = 0.36$, $p = 0.06$). But k is essentially uncorrelated with the lens score ($\rho = -0.03$). Removing k *raises* the correlation to $\rho = 0.840$ ($p = 2 \times 10^{-8}$): k and the lens are independent predictors of faithfulness, so removing the variance attributable to k sharpens rather than weakens the lens signal.[1]

Nor by answer frequency. A relation whose answers are frequent tokens might be easy for both measures. We estimate the model's prior over each answer token by averaging its log-probability across ten neutral contexts. This predicts neither faithfulness ($\rho = -0.02$) nor the lens score ($\rho = -0.01$), and removing it leaves the correlation unchanged at 0.771. The correlation is 0.841 with both cardinality and answer frequency removed.

[1] All variables here are ordinal, so we work throughout with ranks. To remove the effects of a set of covariates, we rank-transform the lens score, faithfulness, and each covariate; regress the ranked lens score and ranked faithfulness separately on the ranked covariates using ordinary least squares with an intercept; and report the Pearson correlation between the resulting residual vectors. This is the standard partial Spearman correlation, and it asks how much of the lens–faithfulness association remains once everything the covariates can linearly explain about either variable has been removed.

Table 1: Object prominence, measured by the model's own free generation and therefore independently of the lens, against how often the lens surfaces the gold object in its top five. Spearman $\rho = 0.614$ ($p = 0.019$, $n = 14$). See §9.2 for discussino of the outlier star_constellation.

relation	prom.	n	gold∈lens	n
city_in_country	1.00	5	0.65	17
person_plays_pro_sport	0.60	40	0.35	20
star_constellation	0.60	40	0.05	20
country_capital_city	0.50	8	0.56	16
landmark_in_country	0.47	40	0.25	20
product_by_company	0.33	40	0.05	20
person_occupation	0.30	40	0.10	20
country_currency	0.29	14	0.00	20
person_university	0.28	29	0.00	20
country_largest_city	0.25	8	0.25	16
person_father	0.13	39	0.06	18
company_hq	0.12	40	0.00	20
superhero_archnemesis	0.05	39	0.05	19
person_mother	0.03	40	0.00	20

Nor by one domain. Eleven of the 28 relations have *person* subjects, raising the worry that one domain (i.e., one subject type) drives the result. Holding out each subject domain in turn leaves $\rho \in [0.570, 0.700]$, median 0.656.

Unchanged findings under an alternative scoring protocol. We reran the analysis in Figure 1 under an alternative protocol: a single shared split and objects restricted to those their first token identifies. A relation meets the restriction if, after discarding every object whose first token it shares with another object of the same relation, enough subjects remain to form a split: eight in-context examples, eight reference subjects and at least one test subject. Three relations fall below that threshold and are dropped – the two with numeric objects, whose years collapse onto shared leading digits, and PERSON_BAND_LEAD_SINGER. Over the remaining 25 the alternative protocol gives $\rho = 0.624$ ($p = 0.0009$), against 0.771 over 28 here. We return to first-token collapse, and to why restricting the gold objects is only a partial remedy, in §9.

7 WHAT THE LENS READS

Our interpretation of Eq. 7 has so far been assumed rather than tested. Readouts of this kind are normally taken to surface what a model is internally disposed to say (Gurnee et al., 2026), and we have read CJL(s) the same way: as what the subject's representation makes available, before any relation has been named. The account of Figure 1 rests on that reading. If it is right, a relation should be linearly decodable to the extent that its answer is already part of what the subject evokes – to the extent, that is, that the object is *salient* for the subject. This section performs a sanity check on that assumption.

Salience, measured independently of the lens. As a check that is external to the lens, we measure salience by the model's own free generation. For each test subject we greedily generate 90 tokens from each of `"{} is best known for"` and `"Here are some facts about {}. First,"`, and record whether any distinctive word of the gold object appears in either continuation.[2] This is a forward pass under different prompts, and involves no Jacobian.

We refer to this rate as the object's *prominence*. Across the 14 relations for which it can be measured it ranges from 1.00 (CITY_IN_COUNTRY) and 0.50 (COUNTRY_CAPITAL_CITY) down to 0.13

[2]Titles, particles and words shared with the subject's own name are excluded, so that *The* Joker is not matched by any occurrence of "the" and *Johann Georg Kant* is not matched by a passage about Immanuel Kant. The lens's own prompt cannot be used here: `"{} is known for"` makes Olmo-3-7B emit a multiple-choice quiz rather than prose, so the object is never named.

(PERSON_FATHER), 0.05 (SUPERHERO_ARCHNEMESIS) and 0.03 (PERSON_MOTHER). It predicts how often the lens surfaces the gold object in its top five: Spearman $\rho = 0.614$ ($p = 0.019$, $n = 14$). Table 1 gives both quantities per relation. The check passes: what the lens surfaces is predicted by a measurement of salience that never consults the lens.

8 OBJECT CARDINALITY

The correlation in Figure 1 is highly significant, but many relations deviate from the trend. We investigate the tails of this distribution, the relations with the largest positive and negative deviations, and find that extreme object cardinality k can partially explain these tails. Concretely, we order the relations by their residual from the least-squares fit in Figure 1; this residual is in units of faithfulness, so $+0.30$ means thirty points more faithful than the lens score predicts.

The largest positive residuals – faithful despite a low lens score – with their cardinality are LANDMARK_ON_CONTINENT ($+0.30$, $k = 189$), NAME_GENDER ($+0.29$, $k = 9.5$) and PERSON_NATIVE_LANGUAGE ($+0.26$, $k = 30.6$). In each, the LRE chooses among a handful of possible answers, and can be faithful without the answer being salient for any particular subject: no name is chiefly "known for" its gender, but there are only two answers to pick between.

The largest negative residuals – unfaithful despite the lens – are PERSON_FATHER (-0.43), PERSON_MOTHER (-0.41) and COMPANY_CEO (-0.32), all with $k \approx 1$ and answer spaces in the hundreds. For these relations, faithfulness is near its floor due to the difficulty of the task and the details of the representation apparently do not change that.

Our account thus holds in the middle of the cardinality range and degrades at both ends, in opposite directions. Cardinality is the same variable whose removal raises the correlation to 0.744 (§6). Our analysis in this section gives that number an interpretation: k is a source of deviation in the salience account rather than merely a nuisance correlate.

9 FACT-LEVEL ANALYSIS

Linearity emerges as a phenomenon at the relation level, not the fact level – it would not make sense to investigate linearity for the facts of a relation that is not represented linearly. But once there is evidence for at least partial linearity of a relation, we can ask for individual facts how well the contrastive Jacobian lens predicts the degree to which they are integrated into the linear structure.

To analyze relations at the fact level, we compute the ρ between lens $\log_{10}$ rank and LRE $\log_{10}$ rank.

In this rank-rank analysis of individual facts for PRODUCT_BY_COMPANY, lens rank and LRE rank are moderately highly correlated: $\rho = .41$ for all points, $\rho = .58$ for misses. This is not the case in general. Ranks are zero-based.

We use this rank-rank setup to investigate fact-level linearity. The analysis shows that lens rank is a moderately good predictor of LRE rank for PRODUCT_BY_COMPANY ($\rho = .41$), especially if we only consider misses ($\rho = .58$). Points with an LRE rank of 0 (i.e., a hit, not a miss) are "censored": They give us no resolution because they fail to distinguish cases where the LRE predicts the correct object perfectly vs those where it barely avoids a miss. So misses may be a better diagnostic for assessing the size of the correlation.

Our original plan was to develop a fine-grained account of what contrastive Jacobian lens readouts predict about the linearity of individual facts. However, we found that several shortcomings of the experimental setup currently used for investigating relations in LMs make this hard. In this section, we analyze these shortcomings and give guidance for future relation datasets that may better support research on how LMs represent facts.

We conduct our analysis on the three relations that have the worst overall correlation in the rank-rank analysis: their correlations are close to zero or even negative.

9.1 SEMANTICALLY WEAK FIRST TOKENS

PERSON_UNIVERSITY and (SUPERHERO_ARCHNEMESIS) exemplify the problem of *semantically weak first tokens*. Many universities have names of the form "University of". Several archnemeses start with "The", "Doctor", "King" and "Lady". So the LM learns to predict the common prefix as the target token for many persons. The standard setup is to count such a prediction as correct. But this is not an effective evaluation: The correct generation of "University" is not a reliable indicator of the factual competence of the LM.

Measuring the collapse rate – the fraction of a relation's distinct objects that share a first token with another – **6 of 40 relations exceed 25%**, led by PERSON_MOTHER (61%), PERSON_FATHER (58%) and PERSON_UNIVERSITY (45%: 29 of 69 objects begin with UNIVERSITY).

Restricting each relation to objects whose first token identifies them repairs this on the gold side. It is only a partial remedy, because faithfulness compares the LRE against the *model's* top-1 (not gold), so what inflates it is the model's own answer being an easy token. In PERSON_UNIVERSITY, after filtering, the model still answers UNIVERSITY for 7 of 29 subjects and the LRE agrees on every one. A complete fix would condition on the prediction being distinctive.

The direction of the bias differs by relation. Where the collapse is on the gold side it makes faithfulness easier – 41 distinct fathers all counting as JOHN – so our PERSON_FATHER figure is an upper bound, and the corrected value is lower still ($0.07 \rightarrow 0.05$). Where the model itself emits the shared token, the collapse manufactures agreement that is not retrieval at all.

We flag all of this because the failure is silent: it produces a perfect or inflated score rather than an obvious error, and it will affect any first-token evaluation whose answers share prefixes – numeric, titled (*Doctor Doom*, *The Joker*) or generic (*University of X*).

Both LRE and the lens work on the token level, the residual stream vector at the last layer, which is focused on producing the correct next token and may not represent reliable information about the tokens to follow. This single-token regime is highly problematic for the standard setup because we are effectively investigating a relation in which all objects have been shortened to their first token. This might be a nonsensical relation even to humans and it is clearly different from the intended relations whose objects are full entity names.

9.2 REFERENCE COMPETITION

One relation is an outlier: STAR_CONSTELLATION has prominence 0.60 yet the lens surfaces the constellation for only 5% of its stars. We suspected an artifact, since a Bayer designation contains the constellation in the Latin genitive (*Eta Persei* for *Perseus*), which our word filter does not remove; but prominence on the 19 subjects without such overlap is 0.63, slightly higher rather than lower, so this explanation fails. What the lens returns for these subjects is category vocabulary – *constellation*, *brightness*, *stellar* – rather than the particular constellation. Excluding the relation would raise the correlation to 0.725.

PERSON_OCCUPATION exemplifies the problem of *subject ambiguity*. An example is Rolf Thommessen who is listed as an actor in LRE. The highest ranked token given by the lens is "whistlebl". This indicates that the most prominent entity associated with the last name Thommessen by Olmo-3-7B is Advokatfirmaet Thommessen, a law firm known for their digital whistleblowing tool. One might hope that the first name "Rolf" disambiguates which subject is intended, but apparently this is not the case. In general, the last few tokens seem to dominate which entity is represented and used by the LM. For the actor Barry Sobel, the lens top-ranks "kosher", "Jewish", "Rabbi", correct associations for the Brazilian Rabbi Henry Sobel. For the actress Vondell Darr, the lens top-ranks "criminal", "murders", "felon", plausible associations for the wrongfully convicted Acacia Lyn-Darr. For the actor Simon Gosejohann, the lens top-ranks "Johann", "Hann", "German", suggesting the LM simply uses the string "Johann" as the representation. While we did not conduct an extensive analysis, it is likely that subject ambiguity is a problem for many relations whose subjects are persons. Many of those persons share the properties that result in Barry Sobel, Vondell Darr and Simon Gosejohann being misrepresented by the model.

9.3 TOWARDS BETTER FACT DATASETS

Our analysis motivates two requirements for future evaluations. First, answer scoring should distinguish complete objects rather than credit a shared, semantically weak first token. Filtering for identifying first tokens provides one diagnostic, but excludes some relations and does not ensure that the model's own prediction is distinctive. Evaluation methods that score complete answers are therefore needed for relations with unavoidable prefix overlap.

Second, a subject name should refer to the intended entity for the model under study. Dataset construction alone may not ensure this, because different models can resolve the same name differently. Evaluations should examine the entity that a model associates with each subject, document ambiguity, and distinguish errors of subject identification from errors in relational decoding. These are methodological directions motivated by the observed failures, rather than solutions evaluated in this study.

10 WHAT THE OPERATOR'S TWO TERMS CARRY

3 <<< hs: add a discussion here of why we drop the anchor Keeping it would instead mean choosing how to weigh it against the remainder, which is what β does in Eq. 3. A single β shared by every relation would be a legitimate option, adding nothing relation-specific; it is simply not a profitable one. In sample such a β slightly outperforms dropping the anchor, but the curve is flat across a wide range of β and the advantage does not survive selection: choosing β on 15 relations and evaluating on the remaining 10, across 2,000 random splits, beats the anchor-free readout in only 30% of them. We therefore drop the anchor, which leaves a readout with no hyperparameter at all – the decoder normalizes, so once the anchor is gone, scaling cannot change a ranking and β has nothing left to do. This gives the *contrastive Jacobian lens* of e,

$$\mathbf{CJL}(e) \;=\; \mathrm{mean}_i \, J_i \, (s_e - s_i), \tag{7}$$

a vector in the final-layer residual stream, which we read out through the model's own output head as a ranking over the vocabulary. >>>

§4 split an LRE into a subject-independent anchor $\mathrm{mean}_i \, f(s_i)$ and a subject-dependent remainder $T_r(e) = \mathrm{mean}_i \, J_i^r (s_e - s_i)$, and dropped the first. This section asks what each term actually carries, by recombining them across relations. It is an intervention rather than a correlation: if the anchor carries the relation, then pairing it with a *foreign* subject term should still produce this relation's object.

Setup. The analysis needs relation pairs meeting two conditions, and only two pairs in the benchmark meet both. The relations must *share subjects*, so that a term can be swapped with nothing else changing; and each must have a *strong operator*, since a term cannot substitute for one that does not produce the answer in the first place. This selects COUNTRY_CAPITAL_CITY and COUNTRY_LARGEST_CITY over 12 countries (faithfulness 1.00 and 1.00), and PERSON_PLAYS_POSITION_IN_SPORT and PERSON_PLAYS_PRO_SPORT over 24 athletes (0.86 and 0.91). Two further candidate pairs fail the second condition and are discussed in Appendix E. Each subject is read with the remaining subjects as references, and in-context demonstrations are drawn from outside the shared set, so no reference or test subject appears in any prompt. We set $\beta = 3$ throughout and report how often the decoded top-1 is the relation's gold object; Appendix E gives the sensitivity to β.

The subject term is interchangeable. Giving an anchor the *other* relation's subject term barely changes the answer (Table 2, column 3): the capital anchor still names capitals 11 times in 12 with the largest-city subject term, and the position anchor still names positions 20 times in 24 with the sport subject term. The anchor decides, not the subject term. This holds across prompts with nothing

		subject term		controls	
relation	n	own	foreign	lens CJL(s)	wrong donor
COUNTRY_CAPITAL_CITY	12	11	11	6	0
COUNTRY_LARGEST_CITY	12	10	9	5	0
PERSON_PLAYS_POSITION	24	19	20	20	5
PERSON_PLAYS_PRO_SPORT	24	22	18	21	3

Table 2: Recombining an LRE's anchor with different subject terms, at $\beta = 3$. Counts are how often the decoded top-1 is the relation's gold object. *Own* is the unmodified operator; *foreign* uses the paired relation's subject term; *lens* uses CJL(s), which never sees a relation. A foreign subject term, and often the relation-agnostic lens, does as well as the operator's own.

in common: a subject term estimated under `"The official currency of {} is the"` still yields capitals 9 times in 12 under the capital anchor.

The lens can replace the subject term. Substituting CJL(s) — estimated under the neutral prompt, with no relation anywhere — for $T_r(e)$ recovers much of the operator (column 4). On the two person relations it matches it: 20 against 19 for position, 21 against 22 for sport. The lens alone never names a sport at all, so what it supplies is the subject and what the anchor supplies is the relation.

Controls. Two nulls establish that something is composed rather than read off the anchor. The *anchor alone* is the $\beta \to 0$ limit; the *wrong donor* pairs an anchor with a different subject's lens vector. For the two city relations both score **zero** at every β we tried, so every hit is attributable to the correct subject's vector. For the person relations the anchor alone is not negligible — 7 and 8, roughly the majority class — but the wrong donor scores *below* it, and the correct lens gains 15 and 18 subjects while losing none (McNemar $p = 6.1 \times 10^{-5}$ and 7.6×10^{-6}).

11 Limitations

The study covers one model and 28 relations. The relationship is correlational: we show that a relation-agnostic readout predicts LRE faithfulness, not that the representation it reads is what the LRE exploits.

The account has a scope condition we can state but not yet explain (§8). It is weakest where the answer space is very small, since an LRE can then be faithful by elimination rather than by reading anything from the subject, and where the answer space is very large and each answer near-unique, since faithfulness is then close to its floor regardless. We show that these are the two tails of a single variable, but we do not have a principled account of where the boundaries lie, and with 28 relations we cannot fit one.

We also probe salience through a single neutral prompt, `"{} is known for"`. That phrasing asks what an entity is *noted* for, which need not coincide with everything its representation carries; attributes such as gender may be well encoded without being something a name is "known for". A different neutral prompt, or an average over several, might track salience more broadly.

Seven of the 28 relations have fewer than ten test items after the in-context and reference subjects are removed, so their faithfulness estimates are coarse (for three of them, quantized to thirds); a rotating-split protocol would sharpen these.

Finally, the diagnostic is defined only for relations whose subjects are entities, which excludes 19 of the dataset's 47 relations.

12 Conclusion

Whether a relation is linearly decodable from a language model is substantially predictable without the relation. A Jacobian lens estimated from subject differences under a neutral prompt, at a fixed layer, ranks the model's own answer highly for exactly those relations whose LREs turn out to be faithful.

The reading we favor is that salience is the mechanism rather than a confound: a relation admits a faithful linear operator to the extent that its answer is already part of what the subject's representation makes the model disposed to say. On that view the wide spread of LRE faithfulness across relations is less a fact about the relations than about the entities they are asked about – and the exceptions are informative, falling at the two ends of the answer-space range where faithfulness can be achieved, or prevented, without the subject's representation mattering much either way.

4 <<< hs: do we explain how the subset of facts was selected that the experiments are done for? for example, i think we're only using 100 from person-occupration? if not: add an appendix section on this and refer to it in the text >>>

AI USE STATEMENT

We used generative AI tools for brainstorming, assisting with software code, and initial drafting and rewriting of parts of the manuscript. All AI-assisted code and text were reviewed and verified by the authors, and generated code was tested for correctness. The authors take responsibility for the final content of this work, including all text, claims, results, and artifacts produced with the aid of generative AI.

REPRODUCIBILITY STATEMENT

We provide sufficient details to reproduce all experiments in the main text and appendix, including the datasets, model checkpoints, prompt templates, evaluation procedures, and hyperparameters used throughout. All experiments use publicly available model checkpoints and datasets. We report the procedures used to estimate linear relational embeddings and the contrastive Jacobian lens, as well as the definitions of all faithfulness and ranking metrics. Additional implementation details, robustness analyses, and per-relation results are provided in the appendix. Code for reproducing the experiments and figures will be released upon publication.

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A BASELINE COMPARISONS

The first two tables report Spearman correlations with LRE faithfulness across the LRE Dataset.

Table 3: **Readout baselines across seven models.** Spearman correlation ρ with LRE faithfulness.

Model	Differential Jacobian	Raw Jacobian	Logit lens	Differential logit lens	Attribute rate
OLMo 3 7B	0.683	**0.689**	0.234	0.326	−0.114
GPT-J 6B	**0.573**	0.567	0.531	0.394	0.002
Mistral 7B	**0.606**	0.586	0.402	0.249	−0.304
Gemma 2 9B	0.554	**0.575**	0.554	0.221	−0.292
Qwen3 8B Base	0.729	0.736	**0.743**	0.607	0.052
Qwen3.8 27B	0.600	**0.644**	0.402	0.454	−0.196
Pythia 6.9B[†]	0.479	**0.501**	−0.343	0.217	0.096

Bold marks the largest point estimate within each model, not a significant difference. [†]Pythia's attribute rate excludes 413 test rows with no predictions surviving the attribute filter; the other models have no undefined rates.

Readout definitions. For subject state s_e, reference states s_i, and reference Jacobians J_i, the differential and raw Jacobian readouts decode $\mathrm{mean}_i\, J_i(s_e - s_i)$ and $\mathrm{mean}_i\, J_i s_e$, respectively. Logit lens decodes s_e; differential logit lens decodes $s_e - \mathrm{mean}_i\, s_i$. Each relation's score is the negative median $\log_{10}(\mathrm{rank} + 1)$ of the model's answer token.

Attribute rate. Following Geva et al. (2023), we measure overlap between filtered top-50 logit-lens predictions and subject attributes retrieved from Wikipedia. The index uses the union of current test subjects; the source stopword, length, and retrieval filters are retained. Relation scores average the percentage overlap. Undefined rates are handled as noted in the table; matched-row sensitivity results remain in the accompanying data.

Raw Jacobian has a larger point estimate than differential Jacobian in five models. All seven paired 95% relation-bootstrap intervals for their difference include zero; these results do not establish a general advantage for either readout. LRE hyperparameter selection remains in sample.

A.1 OLMo CORPUS-FREQUENCY BASELINES

Table 4: **Frequency baselines for OLMo 3 7B.** Correlations with LRE faithfulness across the same 25 shared-split relations, using document counts from OLMo's three published training mixes combined.

Frequency measure	Spearman ρ	p-value
Marginal frequencies		
Subject frequency, c_s	0.290	0.160
Object frequency, c_o	0.644	5.08×10^{-4}
Co-occurrence		
Joint frequency, c_{so}	0.578	0.002
Mean of $\log(1 + c_{so})$	0.568	0.003
$\log(1 + \text{mean}(c_{so}))$	0.578	0.002
Conditional frequencies		
Forward conditional, $P(o \mid s)$	0.386	0.056
Reverse conditional, $P(s \mid o)$	−0.328	0.109
Directional difference, $P(o \mid s) - P(s \mid o)$	0.362	0.075
Micro-averaged $P(o \mid s)$	0.615	0.001
Micro-averaged $P(s \mid o)$	0.030	0.887

Counts denote binary document presence. Means use the current test rows with both marginals observed; micro averages are ratios of summed counts. The p-values are two-sided and unadjusted.

Counting and aggregation. A document contributes at most one mention per subject/object string. Matching uses normalized literal names with Unicode word boundaries, without alias resolution or a proximity constraint. With subject, object, and joint counts c_s, c_o, and c_{so}, the conditionals are $P(o \mid s) = c_{so}/c_s$ and $P(s \mid o) = c_{so}/c_o$. Macro features average row-level quantities where both marginals are observed. Micro conditionals divide summed joint counts by summed marginals. The log of the mean joint count has the same Spearman correlation as the mean joint count because the transformation is monotonic.

Interpretation and reproducibility. These are exploratory associations. Relations can share subjects and objects, and the reported p-values are unadjusted. Conditional direction also reflects marginal frequency: when $c_{so} > 0$, $P(o \mid s)/P(s \mid o) = c_o/c_s$.

A.2 NEUTRAL-PROMPT VARIATIONS

Table 5 collects the prompt experiments for the differential Jacobian lens. For Qwen3 8B Base, we compare the original prompt with six alternatives and the subsequently measured newline control. The checkpoint, layer 19, reference and test rows, relation-prompt answer tokens, and LRE outcomes are fixed within each protocol. The legacy split contains 28 relations; the shared split contains 25. The lower panel reports the original-to-newline comparison for all seven models on the shared split, using each model's fixed layer.

Table 5: **Neutral-prompt variations.** Spearman ρ between differential-Jacobian scores and retained LRE faithfulness. Panel A compares exact templates for Qwen3 8B Base; panel B isolates the newline-prefix comparison across all seven models on the shared split.

A. Qwen3 8B Base: prompt wording and prefix

Neutral prompt	Shared ($n = 25$)	Legacy ($n = 28$)
`{s} is known for`	0.268	0.060
`{s}`	−0.057	0.058
`{s} is`	0.166	−0.118
`{s} can be described as`	0.372	0.179
`Information about {s}:`	0.733	0.295
`Facts about {s}:`	0.788	0.298
`Entity: {s} is known for`	0.728	0.202
`\n{s} is known for`	0.729	0.216

Interpretation. Qwen's shared-split correlation rises from 0.268 under the original prompt to 0.729 with a leading newline; `Facts about {s}:` gives 0.788. These are exploratory sensitivity results, not held-out prompt selection. Prefix wording can change both context and subject position. The newline preserves Qwen's original content token IDs, while Mistral also retokenizes the initial content. The legacy and shared protocols are not independent replications, and no layer or LRE hyperparameter is retuned between prompts.

B JACOBIAN–VECTOR PRODUCTS

Neither the operator nor the lens ever needs J_i itself, only products $J_i u$: for the lens this is explicit in Eq. 7, and for the LRE it is what the expansion in Eq. 3 buys, since it turns $W = \mathrm{mean}_i\, J_i$ into a sum of such products. We obtain each one by the standard double-backward construction (Pearlmutter, 1994): differentiate $\langle u', f(h)\rangle$ with respect to h with u' held as a free variable, then differentiate the result with respect to u' in the direction u. Each product therefore costs a constant number of forward and backward passes rather than the d_{model} passes a materialized Jacobian would need. The construction requires `float32` and eager attention, as the fused attention kernels are not twice differentiable.

C LAYER SELECTION

Layer 17 is roughly half-way through the network, the region in which such lenses are usable at all (Gurnee et al., 2026), and is the modal best layer for the LRE itself in our sweep (10 of 28 relations; the rest spread over 5, 11, 14, 20 and 23). A single layer is used for *every* relation, without per-relation tuning. However, this does not establish robustness to the layer choice: changing the layer can affect relations differently. A systematic sensitivity analysis is left to future work.

D NUMERIC-OBJECT TOKENIZATION CORRECTION

Both relations with numeric objects (PRESIDENTS_BIRTH_YEAR, PRESIDENTS_ELECTION_YEAR) initially scored faithfulness 1.000. This was spurious. In this tokenizer `" 1743"` segments as `[' ', '174', '3']`: digits never merge with a preceding space. First-token scoring therefore compared the model's top-1 against a bare space token on both sides, and faithfulness 1.000 meant only that the LRE and the model both emit a space.

The fix is to place the space in the prompt, so that the query ends with it and the next token is content. Rescored, the two relations give faithfulness 0.67 and lens scores of -4.24 and -4.17; the model does predict the year correctly once the space is consumed. Including the uncorrected values would have depressed the reported correlation from 0.658 to 0.427.

E TERM RECOMBINATION: β SENSITIVITY AND TWO NEGATIVE CASES

5 <<< claude: β is fixed at 3 here, but that value was chosen by inspecting these same subjects, so it is selected on the evaluation set. >>>

Sensitivity to β. Table 2 fixes $\beta = 3$. Table 6 gives the same counts at $\beta \in \{3, 5, 10\}$.

relation	β	own	foreign	lens CJL(s)	wrong donor
CAPITAL_CITY ($n = 12$)	3	11	11	6	0
	5	11	10	8	0
	10	11	8	8	0
LARGEST_CITY ($n = 12$)	3	10	9	5	0
	5	10	9	7	0
	10	8	8	8	0
PLAYS_POSITION ($n = 24$)	3	19	20	20	5
	5	19	16	21	5
	10	17	13	15	2
PLAYS_PRO_SPORT ($n = 24$)	3	22	18	21	3
	5	16	12	22	3
	10	13	3	17	2

Table 6: Counts of Table 2 across β.

The two conclusions hold throughout, but not equally. For the city relations nothing depends on β: the controls stay at zero everywhere and the lens improves with β, from 6 to 8 and from 5 to 8. For the person relations the *foreign subject term* degrades as β grows — sharply for PERSON_PLAYS_PRO_SPORT, from 18 to 3 — because a larger β amplifies the subject term against the anchor until the term's own relation shows through. The lens, by contrast, peaks at $\beta = 5$.

$\beta = 3$ is therefore conservative for the claim about the lens, which does better at 5, and favorable for the claim about foreign subject terms, which do worse above 3. A referee is entitled to note that the second claim is strongest at the value we report.

Two pairs that do not support the analysis. Both fail the strong-operator condition, in different ways, and both are informative about the method's scope.

COUNTRY_CURRENCY shares all 12 countries with the other country relations and has faithfulness 0.36, low but not obviously disqualifying. Its own operator recovers the currency for 6 of 12 countries at $\beta = 3$, 2 at $\beta = 5$ and none at $\beta = 10$. Against that, the lens scores 2, and the anchor alone and the wrong donor both score 3 — the composition is *worse* than its own nulls. Where an operator is this weak there is nothing for a substitute to reproduce, and the counts measure the anchor's own output rather than any composition.

PERSON_FATHER and PERSON_MOTHER are the ideal design on paper: 748 shared subjects, an order of magnitude more than either pair we use, and both answers are person names, so the two relations cannot be told apart by the *kind* of answer — the discriminating case that the country and sport pairs, whose answers differ in kind, do not test. Their faithfulness is 0.06 and 0.12. The operators essentially never produce the gold object, so the experiment cannot be run at all. We note this because it is the one configuration that would test the decomposition in its strongest form, and the benchmark does not currently support it.

F ALL RELATIONS

Table 7 gives faithfulness and the lens score for every relation in the study. Table 1 gives object prominence and the lens's top-five hit rate for the 14 relations on which prominence can be measured (Section 8).

Table 7: All 28 relations, sorted by LRE faithfulness. k is the mean number of subjects per distinct object; A_r is the relation-agnostic lens score, higher meaning closer agreement with the model. †rescored for the numeric-object artifact (Appendix D); *fewer than ten test items, so faithfulness is coarsely quantized.

relation	k	faith.	A_r
country_language*	1.7	1.00	-1.04
country_largest_city*	1.0	1.00	-0.81
landmark_on_continent	189.4	0.94	-2.81
person_native_language	30.6	0.94	-2.56
name_birthplace	3.9	0.93	-1.29
name_religion	6.2	0.93	-1.37
person_plays_pro_sport	63.6	0.92	-1.45
person_plays_instrument	85.5	0.88	-1.77
landmark_in_country	9.2	0.84	-2.25
city_in_country	1.3	0.82	-0.48
person_occupation	26.5	0.76	-2.81
country_capital_city*	1.0	0.75	-0.74
name_gender*	9.5	0.67	-4.67
presidents_birth_year†	1.3	0.67	-4.24
presidents_election_year†	1.1	0.67	-4.17
product_by_company	17.4	0.66	-3.24
person_plays_position_in_sport	68.0	0.64	-3.99
person_band_lead_singer*	1.0	0.60	-1.68
person_university	1.3	0.59	-4.52
company_hq	4.1	0.52	-4.09
superhero_person	1.0	0.48	-2.15
star_constellation	11.7	0.42	-3.67
pokemon_evolutions	1.0	0.39	-3.44
country_currency	1.3	0.36	-4.48
superhero_archnemesis	1.1	0.25	-3.57
person_father	1.0	0.09	-3.69
company_ceo	1.0	0.07	-4.59
person_mother	1.0	0.07	-3.99